Yildiz Technical University

ELECTRICAL – ELECTRONICS FACULTY
CONTROL AND AUTOMATION ENGINEERING DEPARTMENT

KOM4521 MULTIVARIABLE CONTROL THEORY

State Feedback , Observer Design Using Ackermann's Formula and Kalman Filter Comparison in MATLAB

NAME-SURNAME : AYLİN ÖZTÜRK

Signature :

Istanbul, January 2025

Objective

The main objective of this project is to design a state feedback controller and implement both a Luenberger observer and a Kalman filter for a fourth-order system. The project aims to compare the performance of classical observer design with optimal state estimation under measurement noise, and to demonstrate the effectiveness of state feedback in controlling the system dynamics.

Introduction

State-space methods provide a systematic framework for analyzing and designing control systems with multiple states. In this project, a fourth-order system is considered, and its dynamics are represented by matrices A, B, and C.

First, a state feedback controller is designed using Ackermann's formula to place the closed-loop poles at desired locations. Then, a Luenberger observer is implemented to estimate system states based on output measurements. Finally, a Kalman filter is applied to provide optimal state estimation in the presence of process and measurement noise.

The performance of the controllers and observers is evaluated through simulation over a 10-second time interval, and comparisons are made between actual system states, Luenberger observer estimates, and Kalman filter estimates.

1. Methods

1.1. Mathematical Model of the System

The system is modeled as a linear time-invariant (LTI) system with four state variables. Equations 1 and 2 indicate the state- space form.

$$\dot{x} = Ax + Bu \qquad (1)$$

$$y = Cx + Du \qquad (2)$$

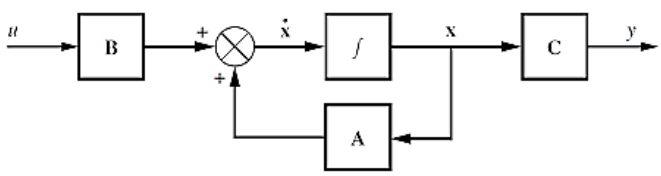

Figure 1: Multi input – multi output system block diagram

Equation 3 gives the dynamical system as a differential equation

$$\frac{d^4\theta(t)}{dt^4} + \alpha^1 \frac{d^3\theta(t)}{dt^3} + \alpha^2 \frac{d^2\theta(t)}{dt^2} + \alpha^3 \frac{d^1\theta(t)}{dt^1} + \alpha^4\, \theta(t) = u(t) \qquad (3)$$

After that, the laplace transform of the system is taken so that we can transform the system into state-space form. We assume that the inital conditions of the states are all zero. Equaiton 4 gives the resulted laplace representation.

$$\theta(s)\left(s^4 + \alpha^1 s^3 + \alpha^2 s^2 + \alpha^3 s^1 + \alpha^4\right) = u(s) \qquad (4)$$

As a result, we get controllable canonical form as stated below A and B matrices. A and B respectively indicate system matrix and input matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\alpha^4 & -\alpha^3 & -\alpha^2 & -\alpha^1 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

1.2. Ackermann's Formula

The purpose of Ackermann's formula is to compute a state feedback gain K such that the closed-loop poles of a controllable LTI system are placed at desired locations. By choosing K appropriately, we can place the closed-loop poles in locations that achieve desired stability, speed of response, and damping. The aim is to design a state feedback controller using Ackermann's formula such that all closed-loop poles are located at -m, where -m is the last digit of the student number (taken as 8).

$$u = -Kx \qquad (5)$$

This equation defines the state feedback control input. Here, uuu is the control signal applied to the system, x is the state vector, and K is the state feedback gain. The negative sign ensures that the feedback stabilizes the system by driving the states toward zero.

New closed-loop system dynamics are obtained as stated in equation 6.

$$\dot{x} = (A - BK)x \qquad (6)$$

$A - BK$ is the closed-loop system matrix. Its eigenvalues (poles) determine the stability and dynamic response of the system. In equation 7, the equation whose roots are the eigenvalues of the closed-loop system is given.

$$\det\left(sI - (A - BK)\right) = \Delta(\mathrm{s}) \qquad (7)$$

Equation 8 states Ackermann's formula for the state feedback gain K.

$$K = [0 \quad 0 \quad 0 \quad 1]\, Cont^{-1}\, \Delta(\mathrm{A}) \qquad (8)$$

First, it is necessary to verify that the given system is controllable. This is done by constructing the controllability matrix given in equation 9.

$$Cont = [B \;\; AB \;\; A^2B \;\; A^3B] \qquad (9)$$

where A and B are the state and input matrices of the system, respectively. If Cont matrix has full rank, then the system is controllable, meaning it is possible to drive the state vector from any initial state to any desired final state in finite time using an appropriate control input. This property is a prerequisite for applying Ackermann's formula, because the formula assumes that the system can be transformed into a controllable canonical form. If Cont matrix is not full rank, state-feedback pole placement is not feasible for the given pair (A,B).

To calculate the desired polynomial, we first select four identical poles located at −m. This yields the polynomial $(s+m)^4$. Next, we replace the variable s with the system matrix A. Then, the resulted desired matrix $\Delta(\mathrm{A})$ is obtained for using in Ackermann formula.

In figure 2, the block diagram of Ackermann's formula is represnted.

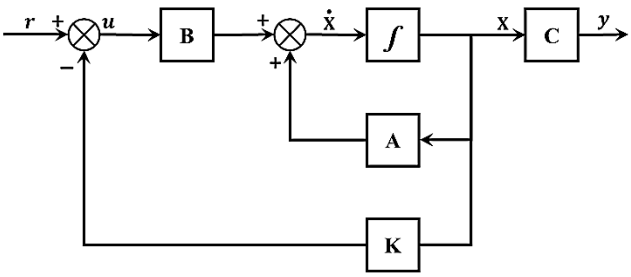

Figure 2: Block diagram of Ackermann's formula

1.3. Luenberger Observer Design

The primary objective of the Luenberger observer is to provide an accurate estimation of the system's internal states in situations where it is not possible to measure all the state variables directly. In practical systems, sensors may only provide measurements of certain outputs, while the remaining states remain unmeasured. The Luenberger observer addresses this limitation by combining the measured outputs $y(t)$ with the system input $u(t)$ and a model

of the system dynamics. It generates an estimated state vector $\hat{x}(t)$ that continuously approximates the true state vector $x(t)$ over time.

Its dynamics are described in equation 10 and 11.

$$\dot{\hat{x}}(t) = \mathrm{A}\hat{x}(t) + \mathrm{B}u(t) + \mathrm{L}\,(y(t) - C\hat{x}(t)) \qquad (10)$$

$$\hat{y}(t) = C\hat{x}(t) + Du(t) \qquad (11)$$

Table 1: Description of parameters of equation 10

$\dot{\hat{x}}(t)$	Estimated state vector
A, B, C	System matrices describing the original system
$u(t)$	Control input
$y(t)$	Measured output
L	Observer gain matrix

Equation 12 and 13 gives estimation error and the observer dynamics from the true system Dynamics

$$\dot{e} = \dot{x}(t) - \dot{\hat{x}}(t) \qquad (12)$$

$$\dot{e}(t) = (A - LC)e(t) \qquad (13)$$

$$e(t) = e^{(A-LC)t}\, e(0) \qquad (14)$$

This shows that the eigenvalues of $A - LC$ determine how fast the estimation error decays. By properly choosing L ,we can place the observer poles at desired locations, ensuring fast and stable convergence of $\hat{x}(t)$ to $x(t)$.

Equation 14 shows that the evolution of the estimation error depends on the eigenvalues of the matrix $A - LC$. By appropriately choosing the observer gain L, the eigenvalues can be placed in the left-half complex plane, ensuring that the estimation error converges to zero as t→∞.

First of all, we need to clarify all states of the system is observable or not via observability matrix which is given in equation 15.

$$O = \begin{bmatrix} \mathrm{C} \\ \mathrm{CA} \\ C^2A^2 \\ C^3A^3 \end{bmatrix} \qquad (15)$$

The system is observable if rank(O)=n and n is the number of states.

We replace our desired observer poles at -3m (m as 8). Equation 16 states Ackermann's formula for the state feedback gain L.

$$L = \Delta(A)\, O^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \qquad (16)$$

Figure 3 illustrates the observer design, where the observer reconstructs the system states from the measured outputs and input signals

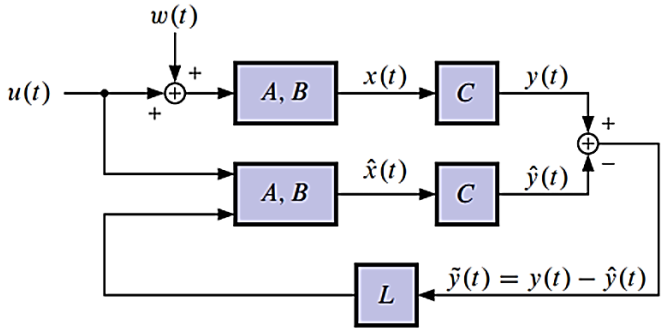

Figure 3.: Closed-loop estimator

1.4. Kalman Filter Implementation

The Kalman filter is a state estimation algorithm that provides optimal estimates of a system's internal states when the system is subject to process disturbances and measurement noise. Unlike the deterministic Luenberger observer, which assumes exact knowledge of the system model and noiseless measurements, the Kalman filter explicitly accounts for uncertainties in both the system dynamics and the sensor measurements. It continuously updates the state estimates by combining predictions from the system model with actual noisy measurements, using a weighted approach determined by the Kalman gain. This gain balances the relative trust in the model versus the measurements, ensuring that the estimation error covariance is minimized over time. As a result, the Kalman filter provides statistically optimal estimates, making it particularly effective in real-world applications where noise and disturbances cannot be ignored. Figure 4 and 5 respectively illustrates discrete-time Kalman filter and continous-time Kalman filter methodology below.

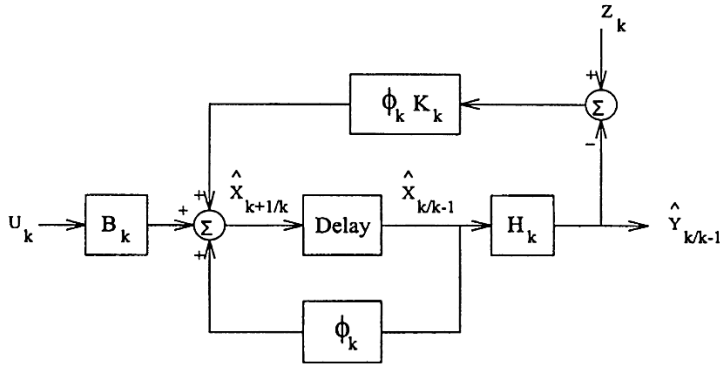

Figure 4.: Discrete-time Kalman-filter block diagram

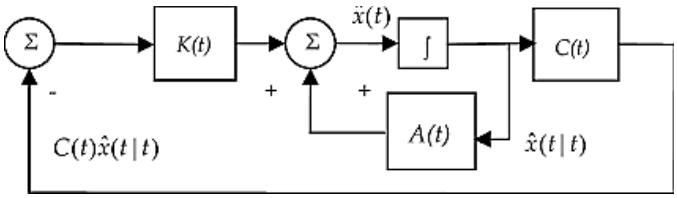

Figure 5.: Continuous-time Kalman-filter block diagram

In this study, the Kalman filter is formulated for a continuous-time system, but in the simulation, the differential equations for the state estimate and error covariance are integrated using a fixed-step Euler method, resulting in a discrete-time approximation

The continuous-time linear system is given in equation 17 and 18.

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t) \qquad (17)$$

$$y(t) = Cx(t) + v((t) \qquad (18)$$

where $x(t)$ is the state vector, $u(t)$ is the system input, $y(t)$ is the measurement, $w(t)$ is process noise with covariance Q, and $v((t)$is measurement noise with covariance R.

The continuous-time Kalman filter estimates the state as in equation 19.

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + K(t)\big(y(t) - C\hat{x}(t)\big) \qquad (19)$$

In equation 20, time-varying Kalman gain is introduced

$$K(t) = P(t)C^T R^{-1} \qquad (20)$$

The error covariance $P(t)$evolving according to the continuous-time Riccati differential equation as stated in equation 21.

$$\dot{P}(t) = AP(t) + P(t)A^T - P(t)C^T R^{-1} CP(t) + Q \qquad (21)$$

In the simulation, these continuous-time differential equations are integrated using a fixed-step Euler method with timestep Δt, yielding a discrete-time approximation:

$$\hat{x}[k+1] = \hat{x}[k] + \dot{\hat{x}}[k]\Delta t \tag{22}$$

$$P[k+1] = P[k] + \dot{P}[k]\Delta t \tag{23}$$

2. Matlab Codes

This section contains the MATLAB scripts used for the analysis and simulation of the control system. The codes include:

I. **State Feedback Design (Part A & B):** Computation of the state feedback gain matrix K using Ackermann's formula and simulation of the closed-loop system

II. **Observer Design (Part C & D):** Design of a Luenberger observer to estimate system states, and simulation of actual vs. estimated state trajectories.

III. **Kalman Filter Comparison (Part E):** Implementation of a Kalman filter to estimate system states in the presence of process and measurement noise, and comparison with the Luenberger observer in the presence of process and measurement noise.

```
%% Aylin Öztürk, 20016928, Department of Control & Automation Engineering
%% PART A
clear all % Clear all variables from the workspace
clc       %Clear the command window

syms  a1 a2 a3 a4 s e  % representing alphas as a symbolic value

m=8; %last digit of my student number
I=eye(4); % determine I matrix - identity matrix 4x4
%state space representation of the system: (dx/dt=A*x+B*u)
A=[0 1 0 0; 0 0 1 0; 0 0 0 1; -a4 -a3 -a2 -a1]; % matrix of system dynamics-A matrix
B=[0;0;0;1]; %input matrix-B matrix

Cont=[B A*B (A^2)*B (A^3)*B]; %our controllbility matrix
if rank(Cont) ~= 4 % checking whether system is controllable or not
    print("system is not controllable"); % printing
    error('error-not controllable'); % if it is not , throw an error
end
```

```

char_equat = (s+m).^4; % desired poles at -m, our characteristic eq.
desired_coeffs_K = coeffs(char_equat, s, 'All') %getting coefficients of char_equat
%constructing the characteristic equation matrix (phi_A) and implement A
%matrix into our phi_A
phi_A =desired_coeffs_K(1)*A^4+ desired_coeffs_K(2)*(A^3) + desired_coeffs_K(3)*(A^2)
+ desired_coeffs_K(4)*A + desired_coeffs_K(5)*I;
K = [0 0 0 1] * inv(Cont) * phi_A; % calculation of K matrix- acakermann formulation for K matrix

disp('state feedback gain matrix K is:');disp(K); % result is displayed

```

```
desired_coeffs_K =

[1, 32, 384, 2048, 4096]

state feedback gain matrix K is:
[4096 - a4, 2048 - a3, 384 - a2, 32 - a1]
```

```
%% PART B
A_Closed_loop= A-(B*K); %closed-loop sys. matrix- new A matrix of closed loop system
x0_states=[m/2;m/2;m;m]; %initial conditions of states are defined
simulation_time=[0:1:10]; %10saniye
x_t = zeros(4, length(simulation_time)); %  x_t matrix to store state trajectories (4 states x time points)

for i = 1:length(simulation_time) % compute state trajectories over time-10sec.
    t = simulation_time(i);%current time
    x_t(:, i) = (expm(A_Closed_loop * t)) * x0_states; %computing states using e^(A_Closed_loop * t) * x0- eA*t
end

% Plot state trajectories
figure;%create a new figure
hold on;%  hold the plot to add multiple curves
plot(simulation_time, x_t(1, :), 'r','LineWidth', 2); %x1(t)-state 1 of the closed loop system -(A-BK)
plot(simulation_time, x_t(2, :), 'g','LineWidth', 2); % x2(t)-state 2 of the closed loop system -(A-BK)
plot(simulation_time, x_t(3, :), 'b','LineWidth', 2); % x3(t)-state3 of the closed loop system -(A-BK)
plot(simulation_time, x_t(4, :), 'k','LineWidth', 2); %x4(t) -state4 of the closed loop system -(A-BK)

xlabel('time in sn');ylabel('state values'); % axis labeling
legend('x1(t)', 'x2(t)', 'x3(t)', 'x4(t)'); % adding legend to plots
title('State Trajectories of the Closed-Loop System (A-BK)'); % title of plot
grid on; %for better visualization
hold off;% elease the hold on the plot

```

```
%% PART C
C= [1 0 0 0]; %output matrix - y=C*x+D*u
Observ=[C;C*(A);C*(A^2);C*(A^3)]; % our observability matrix

if rank(Observ) ~= 4 % checking whether system is observable or not
   print("system is not observable"); % printing error
   error('error-not observable'); % if it is not , system will be throwing an error
end

p_observer = (s + 3*m).^4; % Desired characteristic polynomial for observer
desired_coeffs_observer = coeffs(p_observer, s, 'All'); %getting coefficients of p_observer
%constructing the characteristic equation matrix (phi_A) and implement A
phi_A_obs=desired_coeffs_observer(1)*A^4 +desired_coeffs_observer(2)* A^3 +desired_coeffs_observer(3)*A^2
+desired_coeffs_observer(4)*A +desired_coeffs_observer(5)*I;
L=phi_A_obs*inv(Observ)*[0;0;0;1]; % ackermann formulation for L observer gain matrix

disp('Observer gain matrix L is:');disp(L); % Display observer gain matrix L
A_observer= A-(L*C);
disp(A_observer)
```

```
Observer gain matrix L is:
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```
%% PART B
A_Closed_loop= A-(B*K); %closed-loop sys. matrix- new A matrix of closed loop system
x0_states=[m/2;m/2;m;m]; %initial conditions of states are defined
simulation_time=[0:1:10]; %10saniye
x_t = zeros(4, length(simulation_time)); %  x_t matrix to store state trajectories (4 states x time points)

for i = 1:length(simulation_time) % compute state trajectories over time-10sec.
    t = simulation_time(i);%current time
    x_t(:, i) = (expm(A_Closed_loop * t)) * x0_states; %computing states using e^(A_Closed_loop * t) * x0- eA*t
end

% Plot state trajectories
figure;%create a new figure
hold on;%  hold the plot to add multiple curves
plot(simulation_time, x_t(1, :), 'r','LineWidth', 2); %x1(t)-state 1 of the closed loop system -(A-BK)
plot(simulation_time, x_t(2, :), 'g','LineWidth', 2); % x2(t)-state 2 of the closed loop system -(A-BK)
plot(simulation_time, x_t(3, :), 'b','LineWidth', 2); % x3(t)-state3 of the closed loop system -(A-BK)
plot(simulation_time, x_t(4, :), 'k','LineWidth', 2); %x4(t) -state4 of the closed loop system -(A-BK)

xlabel('time in sn');ylabel('state values'); % axis labeling
legend('x1(t)', 'x2(t)', 'x3(t)', 'x4(t)'); % adding legend to plots
title('State Trajectories of the Closed-Loop System (A-BK)'); % title of plot
grid on; %for better visualization
hold off;% elease the hold on the plot

```

```
%% PART C
C= [1 0 0 0]; %output matrix - y=C*x+D*u
Observ=[C;C*(A);C*(A^2);C*(A^3)]; % our observability matrix

if rank(Observ) ~= 4 % checking whether system is observable or not
   print("system is not observable"); % printing error
   error('error-not observable'); % if it is not , system will be throwing an error
end

p_observer = (s + 3*m).^4; % Desired characteristic polynomial for observer
desired_coeffs_observer = coeffs(p_observer, s, 'All'); %getting coefficients of p_observer
%constructing the characteristic equation matrix (phi_A) and implement A
phi_A_obs=desired_coeffs_observer(1)*A^4 +desired_coeffs_observer(2)* A^3 +desired_coeffs_observer(3)*A^2
+desired_coeffs_observer(4)*A +desired_coeffs_observer(5)*I;
L=phi_A_obs*inv(Observ)*[0;0;0;1]; % ackermann formulation for L observer gain matrix

disp('Observer gain matrix L is:');disp(L); % Display observer gain matrix L
A_observer= A-(L*C);
disp(A_observer)
```

```
Observer gain matrix L is:
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```
[                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```

```matlab
%% D ŞIKKI
% implementing a1=1, a2=1, a3=1, a4=1 and creating new numeric matrises
A_new=[0 1 0 0; 0 0 1 0; 0 0 0 1; -1 -1 -1 -1]; % new numeric A system matrix
phi_A_new =A_new^4+ 32*(A_new^3) + 384*(A_new^2) + 2048*A_new+ 4096*I; %our new A matrix into our characteristic equation
Cont_new=[B A_new*B (A_new^2)*B (A_new^3)*B]; % new numeric controllability matrix
K_numeric = [0 0 0 1] * inv(Cont_new) * phi_A_new; % calculation of new K matrix- acakermann formulation for K matrix
Observ_new=[C;C*(A_new);C*(A_new^2);C*(A_new^3)]; % our observability matrix
phi_A_obs_new= desired_coeffs_observer(1)*A_new^4 + desired_coeffs_observer(2)* A_new^3
+ desired_coeffs_observer(3)*A_new^2 + desired_coeffs_observer(4)*A_new +desired_coeffs_observer(5)*I;
B=[0;0;0;1];

L_new = phi_A_obs_new*inv(Observ_new)*[0;0;0;1]; % ackermann formulation for L observer gain matrix
%combining plant and observer dynamics is A_fuullmatrix:
A_fullmatrix = double([A_new-(B*K_numeric), (B*K_numeric); zeros(size(A_new)), A_new-(L_new*C)]);

x0 = [m/2; m/2; m; m]; % initial values of actual states
x0_estimated = [1; 1; 1; 1]; %initial values of estimated states
x0_full = [x0; x0_estimated]; % containing actual states and estimated states- x0_full

t = 0:1:10;%time vector

x_t_full = zeros(length(x0_full), length(t)); %initialize matrix to store state evolution over time
% i am computing all state trajectories over time -10sec.
for i = 1:length(t)
    x_t_full(:, i) = expm(A_fullmatrix * t(i)) * x0_full;% Solve for states at each time
end

x_t = x_t_full(1:4, :); %extracting actual s from full state trajectory
x_t_estimated = x_t_full(5:8, :); % extracting estimated states from full state trajectory
```

```matlab

% x1 vs x1_estimated
figure; % create new figure for plotting
hold on; %hold the plot to add multiple curves
plot(t, x_t(1, :), 'b','LineWidth', 2); % plot actual x1 trajectory
plot(t, x_t_estimated(1, :), '--r','LineWidth', 2);% plot estimated x1 trajectory
xlabel('time in sec.');ylabel('x1 & x1_estimated of overall system');%axis labeling
legend('x1(t)', 'x1(t)estimated'); %adding legend
title('actual and estimated state trajectories of the overall closedloop system - x1(t)'); %title of graphic
grid on;
hold off;

% x2 vs x2_estimated
figure;
hold on;
plot(t, x_t(2, :), 'b','LineWidth', 2); % plot actual x2 trajectory
plot(t, x_t_estimated(2, :), '--r','LineWidth', 2); %plot actual x2 trajectory
xlabel('time in sec.');ylabel('x2 & x2_estimated');%axis labeling
legend('x2(t)', 'x2(t)estimated of overall system');%adding legend
title('actual and estimated state trajectories of the overall closed loop system -x2(t) ');%title of graphic
grid on;hold off;

% x3 vs x3_estimated
figure;
hold on;
plot(t, x_t(3, :), 'b','LineWidth', 2);% plot actual x3 trajectory
plot(t, x_t_estimated(3, :), '--r','LineWidth', 2); %plot actual x3 trajectory
xlabel('time in sec.');ylabel('x3 & x3_estimated');%axis labeling
legend('x3(t)', 'x3(t)estimated of overall system');%adding legend
title('actual and estimated state trajectories of the overall closed loop system -x3(t)');%title of graphic
grid on;hold off;

```

```matlab

% x4 vs x4_estimated
figure;
hold on;
plot(t, x_t(4, :), 'b','LineWidth', 2); % plot actual x4 trajectory
plot(t, x_t_estimated(4, :), '--r','LineWidth', 2); % plot actual x4 trajectory
xlabel('time in sec.');ylabel('x4 & x4_estimated of overall system'); %axis labeling
legend('x4(t)', 'x4(t)estimated');%adding legend
title('actual and estimated state trajectories of the overall closedloop system -x4(t)');%title of graphic
grid on;hold off;

```

```
%% E) Observer Design under Noise vs Kalman Filter
A_obs = A_new;
B_obs = B;
C_obs = C;

% Noise covariance
Q = 0.01*eye(4);  % process noise
R = 0.1;          % measurement noise

dt = 0.01;
Tend = 10;
t = 0:dt:Tend;
n = length(t);

% Initial conditions
x_true    = zeros(4,n);  x_true(:,1) = [m/2; m/2; m; m];
xhat_L    = zeros(4,n);  xhat_L(:,1) = [1;1;1;1];      % Luenberger
xhat_KF   = zeros(4,n);  xhat_KF(:,1) = [1;1;1;1];      % Kalman
P         = eye(4);
u         = zeros(1,n);


p_observer = (s + 3*m)^4;
desired_coeffs_observer = coeffs(p_observer, s, 'All');
phi_A_obs=desired_coeffs_observer(1)*A_obs^4 +desired_coeffs_observer(2)* A_obs^3 ...
    +desired_coeffs_observer(3)*A_obs^2 +desired_coeffs_observer(4)*A_obs ...
    +desired_coeffs_observer(5)*eye(4);
Observ = [C_obs; C_obs*A_obs; C_obs*A_obs^2; C_obs*A_obs^3];

rank_O = rank(Observ);

if rank_O == rank(A)
    disp('System is observable');
else
    disp(['System is NOT fully observable. Rank = ', num2str(rank_O)]);
end
```

```
L_new = phi_A_obs*inv(Observ)*[0;0;0;1];  % Luenberger gain

for k = 1:n-1
    % Real system
    dx = A_obs*x_true(:,k) + B_obs*u(k);
    x_true(:,k+1) = x_true(:,k) + dx*dt;

    % Measurement with noise
    y_meas = C_obs*x_true(:,k+1) + sqrt(R)*randn;

    % Luenberger observer update
    dxhat_L = A_obs*xhat_L(:,k) + B_obs*u(k) + L_new*(y_meas - C_obs*xhat_L(:,k));
    xhat_L(:,k+1) = xhat_L(:,k) + dxhat_L*dt;

    % Kalman filter update
    Pdot = A_obs*P + P*A_obs' - P*C_obs'*(R\C_obs)*P + Q;
    P = P + Pdot*dt;
    K_kf = P*C_obs'/R;
    dxhat_KF = A_obs*xhat_KF(:,k) + B_obs*u(k) + K_kf*(y_meas - C_obs*xhat_KF(:,k));
    xhat_KF(:,k+1) = xhat_KF(:,k) + dxhat_KF*dt;
end

% Plot comparison
state_labels = {'x1','x2','x3','x4'};
for i = 1:4
    figure; hold on; grid on;

    plot(t, xhat_L(i,:), '-g', 'LineWidth', 7);     % Luenberger with noise
    plot(t, x_true(i,:), 'k-', 'LineWidth', 3);       % Actual
    plot(t, xhat_KF(i,:), '--r', 'LineWidth', 2);   % Kalman

    xlabel('Time [s]'); ylabel(state_labels{i});
    legend('Luenberger','Actual','Kalman','Location','best','AutoUpdate','off');
    title(['Observer Performance under Noise - ', state_labels{i}]);
    hold off;
end
```

3. Results and Discussions

Figure 1 shows the state trajectories of the closed-loop system using the state feedback gain matrix K. All states converge smoothly to zero, demonstrating that the system is stable under the designed state feedback control. It also indicates that the system is asymptotically stable

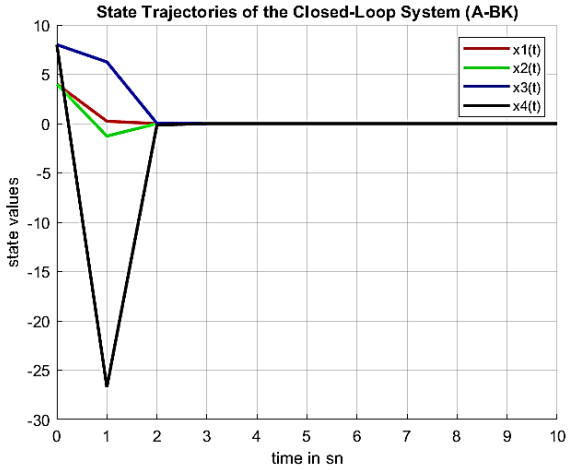

Figure 5.: State trajectories of the closed loop system

As shown in Figures 2, 3, 4, and 5, the Luenberger observer was able to estimate all system states accurately when no noise was present. The estimated states closely matched the actual states, and the observer converged quickly, typically within 2 seconds. These results indicate that the observer design is effective under ideal conditions

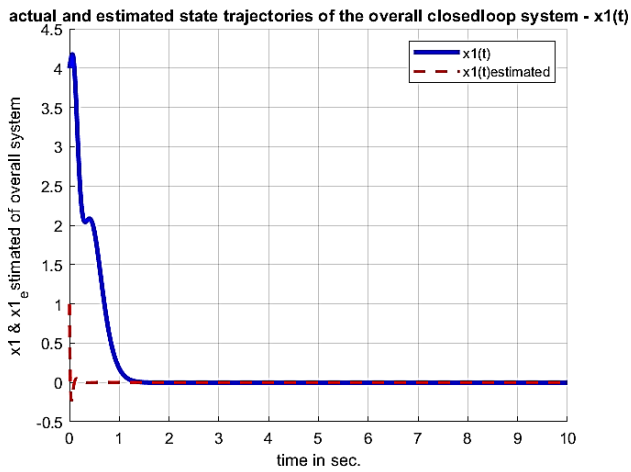

Figure 6.: Comparison of actual and estimated state (x1) with Luenberger observer in noise-free environment

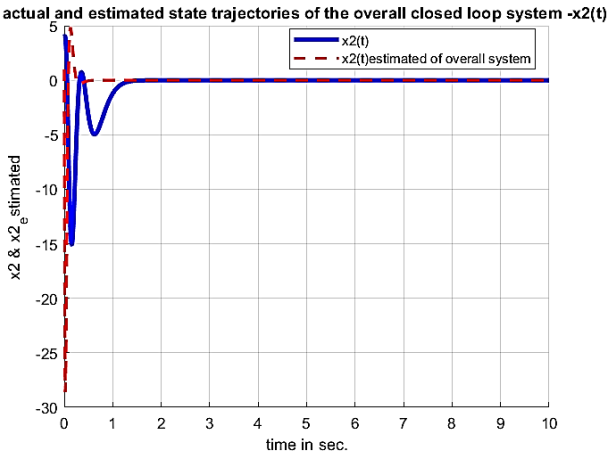

Figure 7.: Comparison of actual and estimated state (x2) with Luenberger observer in noise-free environment

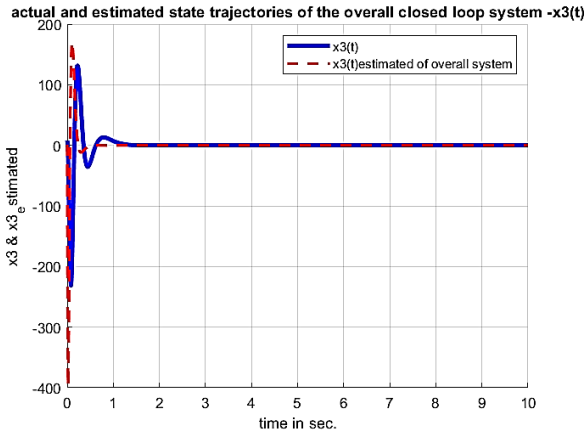

Figure 8.: Comparison of actual and estimated state (x3) with Luenberger observer in noise-free environment

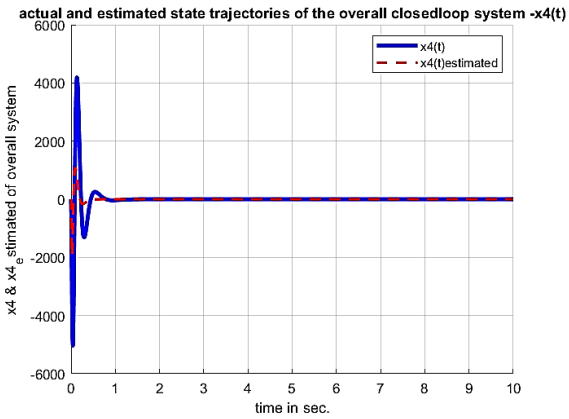

Figure 9.: Comparison of actual and estimated state (x4) with Luenberger observer in noise-free environment

As shown in Figures 6, 7, 8, and 9, the Kalman filter maintained accurate and smooth state estimation even in the presence of noise. It effectively filtered out high-frequency disturbances and provided stable estimates for all states. These results demonstrate that the Kalman filter is more robust than the Luenberger observer under realistic, noisy conditions.

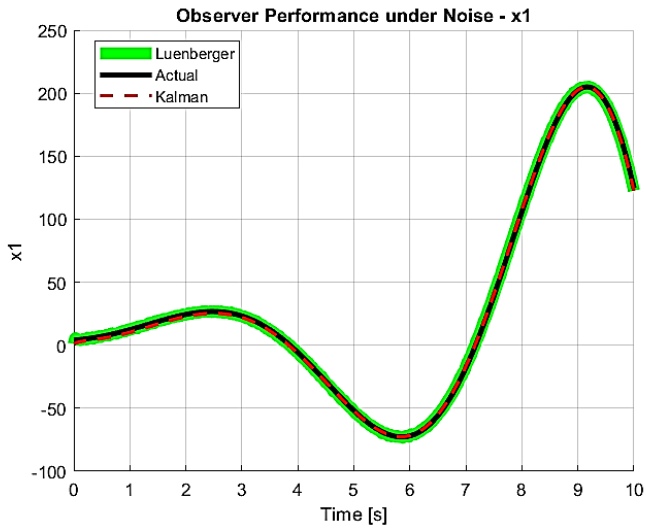

Figure 10.: Comparison of Luenberger observer and Kalman Filter in the presence of noise for x1 state

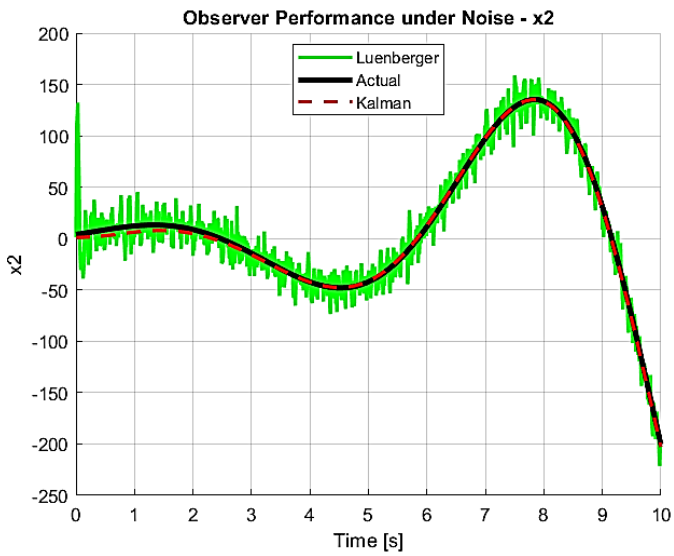

Figure 11.: Comparison of Luenberger observer and Kalman Filter in the presence of noise for x2 state

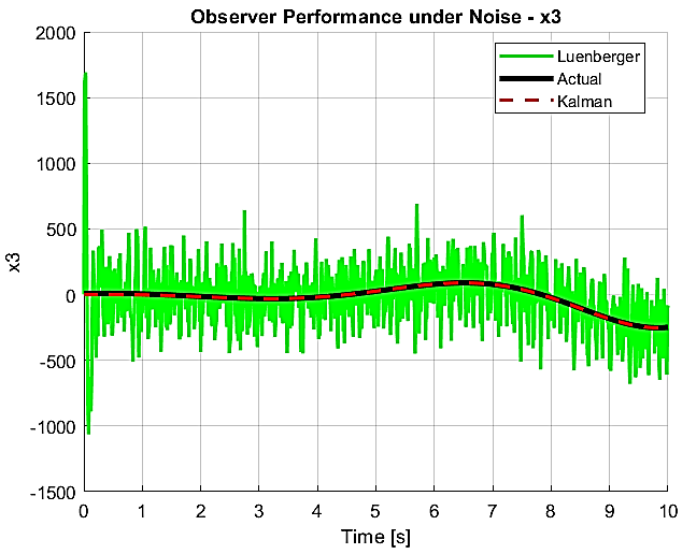

Figure 12.: Comparison of Luenberger observer and Kalman Filter in the presence of noise for x3 state

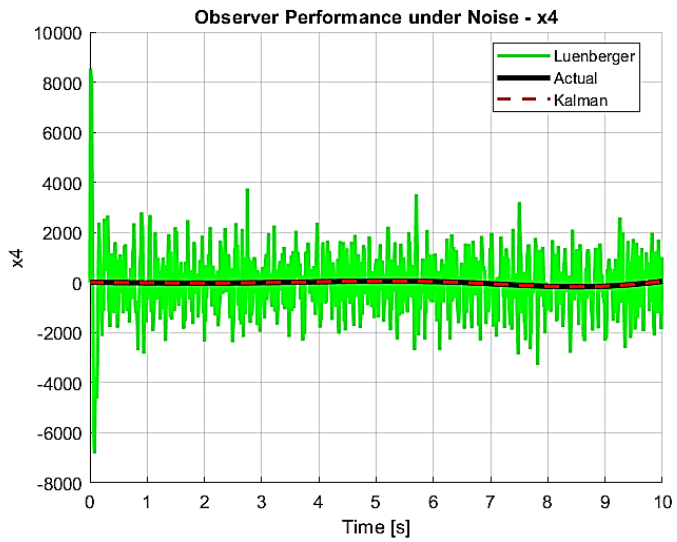

Figure 13.: Comparison of Luenberger observer and Kalman Filter in the presence of noise for x4 state

The results clearly highlight that while the Luenberger observer works well in noise-free environments, the Kalman filter provides more reliable performance in realistic scenarios with measurement disturbances.

4. Conclusions

The state feedback design using the calculated gain matrix successfully stabilizes the system, as confirmed by the state trajectories. All system states converge smoothly to their desired values over time, indicating that the closed-loop system is stable and that the controller is effective in regulating the system dynamics.

The comparative analysis between the Luenberger observer and the Kalman filter revealed distinct strengths under varying noise conditions. In the noise-free scenario, the designed observer exhibited rapid convergence, successfully estimating the true states within approximately two seconds, indicating that the chosen gain matrix provided sufficient dynamic performance for accurate reconstruction of the system states.

However, when process and measurement noise were introduced, the performance of the Luenberger observer deteriorated, exhibiting noticeable fluctuations and reduced estimation accuracy. In contrast, the Kalman filter maintained stable performance in the presence of noise, effectively filtering out high-frequency disturbances and providing smooth, accurate state estimates. This robustness can be attributed to the Kalman filter's statistical optimization framework, which explicitly accounts for noise characteristics through covariance modeling.

Overall, the study emphasizes the importance of considering noise and uncertainties when designing observers for practical systems. While the Luenberger observer is efficient for ideal scenarios, the Kalman filter proves to be more suitable for real-world applications, offering better robustness and reliability in noisy conditions. These findings can guide engineers in choosing appropriate observer designs for control systems operating under different environmental conditions.